

Zain Khan

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EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Applied Science (BASc), Computer Engineering

Sep. 2025 – Apr. 2030 (Expected)

– Over **\$40,000+** in merit-based scholarships, including the **University of Waterloo Presidential Scholarship**.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C/C++, Java, C#, HTML/CSS, SQL

Frameworks & Libraries: React, Next.js, Node.js, Express.js, FastAPI, Tailwind CSS, Chart.js

Databases & Backend: PostgreSQL, MongoDB, Supabase, Firebase

Tools & Platforms: Git, Docker, Linux, Vercel, GitHub API, Postman, Figma

Concepts: System Design, API Design, DSA, OOP, Concurrency, Performance Optimization, Auth, Caching

EXPERIENCE

WATonomous

Waterloo, ON

Robotics Software Engineer — EVE Autonomy

Jan. 2026 – Present

- Developed **concurrent autonomous robotics software in C++ (ROS 2 Humble)**, coordinating **5+ real-time nodes** across perception, planning, and control.
- Designed **modular ROS 2 node architecture** spanning perception, world modeling, planning, and control to enable isolated testing in simulation.
- Implemented **costmaps, A* path planning, and Pure Pursuit control**, achieving stable point-to-point navigation in cluttered environments.

Kasperia

Remote

Software Developer (Frontend & Data) — Contract

Jun. 2023 – Sep. 2025

- Improved page load performance by **30%** and reduced maintenance overhead by **25%** through reusable React components and modular CSS.
- Built **automated data pipelines** using JavaScript, SQL, and REST APIs, reducing manual reporting time by **40%**.
- Developed KPI dashboards with React and Chart.js, increasing stakeholder engagement by **35%**.

PROJECTS

otto  | *Next.js, TypeScript, Python, LiveKit WebRTC, Deepgram, Gemini 2.5 Flash, Supabase*

- Built an **AI voice agent** automating **GitHub, Gmail, and Google Calendar** workflows via natural speech; **1st Place — LiveKit Track @ NexHacks**.
- Implemented a **LiveKit WebRTC pipeline** with **Deepgram VAD** turn detection and **Gemini 2.5 Flash Realtime** native audio inference.
- Integrated **OAuth 2.0 auto-refresh** using **Supabase Auth** for secure, uninterrupted voice commands across services.
- Compressed contextual payloads using **TTC Bear-1**, reducing tokens **4000 → 1800 (55%)** for lower latency and API cost savings.

Sketchy  | *Next.js, TypeScript, Gemini 2.5 Flash, tldraw, MongoDB, GitHub OAuth, Vercel API*

- Built a sketch-to-site web app converting hand-drawn wireframes into deployable HTML/CSS/JS in **under 60 seconds** end-to-end.
- Optimized a **Gemini 2.5 Flash inference pipeline** with constraint-driven prompting; reduced tokens **3000 → 800 (73%)**.
- Implemented an **end-to-end CI-style deployment pipeline** integrating GitHub and Vercel APIs to generate repos, push source, and return live URLs.
- Added **GitHub OAuth** authentication and a **MongoDB Atlas** persistence layer supporting autosave, versioning, and rollback.

inven[s]tory  | *React, Next.js, TypeScript, Supabase, PostgreSQL, Tailwind*

- Developed a **personal inventory and P2P exchange platform** with authenticated users and item lifecycle workflows.
- Designed **normalized PostgreSQL schemas** with Supabase-backed authentication and fine-grained access controls.